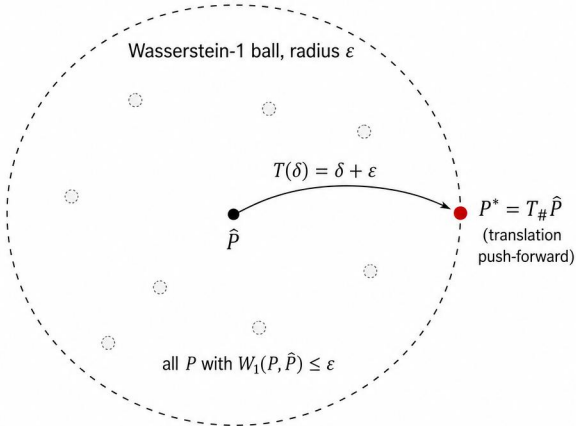




$$\mathbb{E}_{\hat{P}}[\text{arr}(c)] \leq \mathbb{E}_P[\text{arr}(c)] \leq \mathbb{E}_{\hat{P}}[\text{arr}(c)] + \varepsilon$$

$$\text{LCB}(c) = \mathbb{E}_{\hat{P}}[\text{arr}] + \varepsilon = \sup_{P: W_1(P, \hat{P}) \leq \varepsilon} \mathbb{E}_P[\text{arr}(c)]$$

upper bound $\leq$: weak Kantorovich–Rubinstein

lower bound $=$: explicit translation witness P^*